WAVE-DRIVEN ROTARY ENERGY GENERATION SYSTEM

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Project Group No: ECS_36

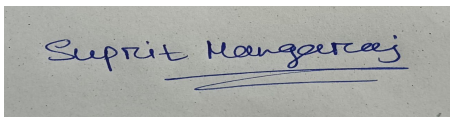
Abstract: The objective of this project is to develop a micro-scale wave energy conversion system that captures the linear motion of water waves and transforms it into rotational kinetic energy to drive a DC generator. The system integrates a modular wave chamber with a turbine unit and utilizes IoT-based monitoring to track real-time power generation metrics. This work aims to provide a sustainable, low-cost energy solution for off-grid coastal applications.

Individual contribution and findings: I contributed to the core electronics development and integration of the system. I worked on the piston mechanism, ensuring proper actuation and synchronization for effective wave generation. I was responsible for setting up and interfacing the multimeter for accurate current measurement. I calibrated the electrical readings to ensure consistency and reliability in the data. I also verified circuit connections and optimized performance during operation. I assisted in troubleshooting issues related to motor control and measurement setup. My contribution ensured stable electronic operation and accurate data acquisition for the system.

Individual contribution to project report preparation: My contribution to the project report preparation focused on documenting the core electronics concepts and system working. I explained the piston-based actuation mechanism and its integration with the motor driver circuit. I detailed the multimeter-based current measurement setup and calibration process. I also ensured clear representation of circuit operation and overall system functionality in the report.

Individual contribution for project presentation and demonstration: My contribution is in preparing and presenting the core electronics and working mechanism of the system. During the demonstration, I explained the piston-based wave generation and showcased the measurement of electrical output using the multimeter.

Full Signature of Supervisor/s: Full signature of the student:



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